

Optimization Project Proposal Winter'21

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I. LITERATURE REVIEW

Multi-robot systems (MRS) employ the idea that there is strength in numbers. The advantages of using multiple robots to perform a certain task become especially noticeable when the task at hand requires multiple simultaneous operations. In this task, the restrictions of a single robot system can be overcome by MRS. In several applications such as exploration, a large number of simple and cheap robots could have better performance than an individual, highly specialized, and costly robot. Another advantage of MRS is redundancy as tasks can be reallocated to different robots in case of failure of one or more of the robots. This can increase the reliability of the system.

There are different approaches to achieve distance-based formation. A genetic algorithm (GA) to find a suitable solution based on angle, distance, and speed to avoid clashing is presented in [1]. Then an extension to a parallel scheme is done in order to improve its performance. This strategy allows robots to share their solutions, improve them, and accelerates convergence time for all robots. Therefore, it can be acknowledged as Distributed Artificial Intelligence. The approach is tested in numerical simulations and real-time experiments in which, through the migration of the solution, the robots share the best paths to converge on the desired distances and avoid collisions.

One of the advantages of optimization is the ability to optimize for a different number of objectives and choosing the trade-offs that better suit the application. The grey wolf optimization technique is used in [2] to create a map of indoor spaces. The exploration problem in their case has two conflicting objectives: searching for new places and enhancing the already explored map accuracy.

The exploration is simulated in a grid where robots emit a sensor beam onto grid cells to mark the cells as explored. However, the sensor beam is not always identical which introduces probability values for each explored cell. The objective of the optimization function is to maximize the number of explored cells and minimize the probability values of all explored cells. The proposed multi-objective grey wolf optimizer is tested in simulation under the constraints of the number of waypoints and iterations. The results show that it outperformed deterministic exploration in all sets of constraints.

One of the heuristic search technique that is considered relatively new, simple, and fast converging is the Particle Swarm Optimization technique (PSO). It is inspired by the behavior of flocks of birds. It is an effective alternative for finding a solution for scheduling problems especially by

utilizing population-based features with it. This algorithm is used by [3] to solve the well-known problem of task allocation in a warehouse. The problem can be summarized as follows, there is a certain number of AGVs, a grid map 2x2, a set of tasks and a certain amount of shelves. The shelves are considered as obstacles. The task allocation problem in this study handles dynamic customer demands which means that new orders can come at any time. This requires the solution to be generated fast enough to run in real-time and take decisions upon it. However, the collision between AGVs is not handled. It is required to minimize the maximum completion time of the given set of tasks. The results are compared with those coming from solving the same problem, but by using a genetic algorithm (GA). The results show that PSO is far more superior to the GA.

Flocking problem is one of the very familiar problems studied in different fields. Flocking behavior is coming from nature such as swarming of bacteria, and biochemical cellular networks. In this problem, three main concepts are achieved: avoiding collision, matching velocity, and staying nearby flockmates. The simulated annealing algorithm is used in [4] to solve the flocking control problem of mobile robots in an environment occupied with unknown obstacles. The distance between two robots is reduced by finding the quasi-optimal position of robots.

In the practical implementation, the mobile robot "AmigoBot" is used with sonar sensors, and a model to explain the flocking behavior is implemented. Also, the kinematic modeling of the mobile robot is constructed assuming no slippage in the wheels of the mobile robot, and controlled using only the two velocities of the wheel motors. The potential function of coordination, as well as the potential function of obstacle avoidance are constructed to test the simulated annealing algorithm. The robot can identify obstacles using sonar sensors during each sample time (500 ms). In addition, the quasi-optimal direction with SA algorithm can be acquired by the agent.

The formal problem of localization of the extrema of a scalar physical field is discussed in [5] and an advanced cat swarm optimization technique is proposed to solve the previously mentioned problem. The problem is formulated by assuming a 2D search space and the authors are concerned only with searching the space not taking into consideration obstacles or collisions. The robots consist of mother robots. Mother robots carry a fixed number of non-returnable satellite devices (child robots). These devices can only measure the value of the scalar field once at a distance that does not exceed a certain value. This means that each mother robot can only

perform a limited number of remote tests. Mother robots can also perform tests at their stop position. The implementation of the ACSO on the problem was successful and showed high efficiency.

II. INTELLIGENT LOGISTIC SOLUTION FOR OPERATION IN WAREHOUSES

The logistic sector is shaping the future of industries. Many enterprises are shifting to the adoption of more and more intelligent logistic solutions either for indoor or outdoor usages. These solutions are helping them in increasing the efficiency of production and efficiency of delivery to customers. On the other hand, the pandemic has imposed high pressure on the logistic sector in order to meet the high demands during and after the lockdown period.

Here, intelligent logistic solutions come to play an important role in warehouses by dispatching goods autonomously using multi-cooperative mobile robots. This idea is supported by the work done in [3]. The idea is to use a number of autonomous mobile robots to handle imports and exports from the warehouse so that we can optimise the dispatching time. The warehouse will consist of a free area for the autonomous mobile robots to move on, the initial position for all the mobile robots (parking area), storage locations (shelves) that contain goods and products, and picking stations that will receive goods and products from the mobile robots for packaging and dispatching. The storage locations will be considered as obstacles for the mobile robots. Our main target is to assign a sequence of tasks for each mobile robot so that they work together taking into consideration: avoiding collisions and in the same time acquiring the optimal path for all the robot to minimize the required time for the dispatching process which will reduce the process time, that will result in better satisfaction for the customer and eventually increasing the profit of the business.

Keywords—logistics, multi-agent, cooperative, autonomous, mobile-robot, optimization.

III. FLOCKING CONTROL OF MOBILE ROBOTS

With the recent construction development, parts are being shipped to the site already fully built and just assembled. That is a better alternative to being constructed on-site as it can save so much hassle. However, with such heavy and big objects the need for a better solution to carry them arises. Like a flock of ants that need to help each other to carry heavy material, the same behavior can be helpful in the desired application. Here, Flocking Control of Mobile Robots comes in handy to perform such a task. Another application for flocking is keeping an optimal distance between flying UAVs that are part of ad-hoc networks. Communication between several flying vehicles is made possible at specific distances which can be achieved by flocking.

The flocking behavior is supported by the work done in [4] it involves multiple autonomous vehicles moving from an initial position to a goal point. Its objectives are keeping the distance between every two flockmates in a predetermined range. This

intern will keep the flock centered. Another objective is matching the velocity of the entire flock. At the same time, each vehicle is capable of avoiding unknown obstacles through sonar sensors that can determine the position of the obstacle.

Keywords—flocking, construction, autonomous, collision-avoidance, optimization, meta-heuristic

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